

Test Cases Document

Project

Community Library Management System

Overview

This document contains the manual test cases prepared for the Community Library Management System. The purpose of these tests is to verify that the application behaves correctly under normal conditions, invalid inputs, and edge cases.

The test cases cover all major modules of the application, including:

- Member Management
- Book Management
- Borrowing and Returning
- Fine Management
- Reports
- Error Handling and Validation

Each test case includes:

- Objective
 - Sample Inputs
 - Preconditions
 - Steps to execute
 - Expected result
 - Pass/Fail criteria
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1. Member Management Test Cases

Add Member with Valid Data

Objective

Verify that a member can be added successfully using valid details.

Sample Inputs

- Name: Ravi Kumar

- Phone: 9876543210
- Email: ravi.kumar@gmail.com
- Membership Type: Basic

Expected Behavior

- Member should be created successfully
 - Member ID should be generated automatically
 - Default account status should be Active
-

Add Member with Duplicate Phone Number

Objective

Verify that duplicate phone numbers are not allowed.

Sample Inputs

- Name: Arun Raj
- Phone: 9876543210
- Email: arun@gmail.com

Expected Behavior

- System should reject the operation
 - Proper validation or database error message should be shown
 - No duplicate record should be inserted
-

Search Member by Phone

Objective

Verify that members can be searched using phone numbers.

Sample Inputs

- Phone: 9876543210

Expected Behavior

- Correct member details should be displayed
-

Search Member by Email

Objective

Verify email-based member lookup.

Sample Inputs

- Email: ravi.kumar@gmail.com

Expected Behavior

- Existing members should be displayed correctly
 - Non-existing emails should show a suitable message
-

2. Book Management Test Cases

Add Book with Existing Category

Objective

Verify adding a book using an already existing category.

Sample Inputs

- Category: Fiction
- ISBN: 9780451524935
- Title: 1984
- Author: George Orwell
- Edition: 1st Edition
- Price: 299.99

Expected Behavior

- Book should be inserted successfully
 - Existing category should be linked properly
-

Add Book with New Category

Objective

Verify automatic category creation when the category does not exist.

Sample Inputs

- Category: Mystery
- ISBN: 9780061122415
- Title: To Kill a Mockingbird
- Author: Harper Lee
- Price: 399.50

Expected Behavior

- New category should be created
 - Book should be linked to the newly created category
-

Add Book Copy

Objective

Verify adding physical copies for a book.

Sample Inputs

- Barcode: BC1001
- ISBN: 9780451524935

Expected Behavior

- Copy should be created successfully
 - Default status should be Available
-

Search Book by Title

Objective

Verify title-based book search.

Sample Inputs

- Title: 1984

Expected Behavior

- Matching books should be displayed with complete details
-

Search Book by Author

Objective

Verify author-based book search.

Sample Inputs

- Author: George Orwell

Expected Behavior

- All books written by the specified author should appear
-

3. Borrowing and Returning Test Cases

Borrow Available Book

Objective

Verify successful borrowing of an available book copy.

Sample Inputs

- Member ID: 2
- Barcode: BC1001

Expected Behavior

- Borrowing transaction should be created
 - Book copy status changes to Borrowed
 - Due date should be generated automatically
-

Borrow Beyond Membership Limit

Objective

Verify borrowing limit enforcement.

Sample Inputs

- Member ID: 2
- Barcode: BC2001

Expected Behavior

- Additional borrowing should be blocked
 - User should receive appropriate message
-

Return Book on Time

Objective

Verify successful return before due date.

Sample Inputs

- Barcode: BC1001

Expected Behavior

- Borrowing status changes to Returned
 - Book copy becomes Available again
 - No fine should be generated
-

Return Book Late

Objective

Verify fine generation for overdue returns.

Sample Inputs

- Barcode: BC3001
- Return Date: 2026-05-20

Expected Behavior

- Late fine should be created automatically
 - Fine amount should be calculated correctly
-

4. Fine Management Test Cases

View Pending Fines

Objective

Verify viewing unpaid fines for a member.

Sample Inputs

- Member ID: 2

Expected Behavior

- All pending fines should be listed correctly
-

Pay Fine

Objective

Verify fine payment functionality.

Sample Inputs

- Fine ID: 1
- Amount Paid: 50.00

Expected Behavior

- Payment should be recorded
 - Fine status should become Paid when fully settled
-

5. Reports Test Cases

Available Books by Category

Objective

Verify category-wise availability report.

Sample Inputs

- Category ID: 1

Expected Behavior

- Available books for the selected category should be displayed
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Member Borrowing History

Objective

Verify member borrowing history retrieval.

Sample Inputs

- Member ID: 2

Expected Behavior

- Both active and returned borrowings should be shown
-

6. Validation and Error Handling Test Cases

Invalid Member ID

Objective

Verify handling of invalid member IDs.

Sample Inputs

- Member ID: 99999

Expected Behavior

- System should show proper error messages
 - Application should continue running safely
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Invalid Barcode

Objective

Verify handling of invalid book copy barcodes.

Sample Inputs

- Barcode: INVALID999

Expected Behavior

- Borrowing or return operation should fail safely
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Empty Required Fields

Objective

Verify validation of mandatory inputs.

Sample Inputs

- Name: ""
- Phone: ""
- Email: ""

Expected Behavior

- System should prevent invalid submissions
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Invalid Numeric Input

Objective

Verify handling of incorrect numeric inputs.

Sample Inputs

- Fine ID: ABC
- Amount: XYZ

Expected Behavior

- Parsing or validation errors should be handled gracefully
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Test Execution Recommendation

Recommended execution order:

1. Member Management
2. Book Management
3. Borrowing and Returning
4. Fine Management
5. Reports
6. Error Handling

This sequence helps create the necessary data before dependent workflows are tested.

Success Criteria

The testing phase will be considered successful if:

- Core workflows operate correctly
 - Business rules are enforced properly
 - Reports display accurate information
 - Invalid operations are handled safely
 - Application does not crash during incorrect input scenarios
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Conclusion

These test cases validate the major workflows and business rules of the Community Library Management System. The testing approach focuses on functional correctness, data consistency, user input validation, and reliable transaction handling across all major modules of the application.